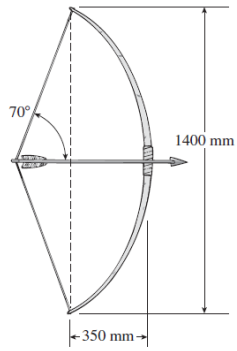
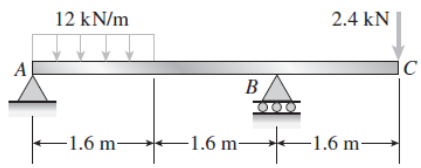
Bending of Beams

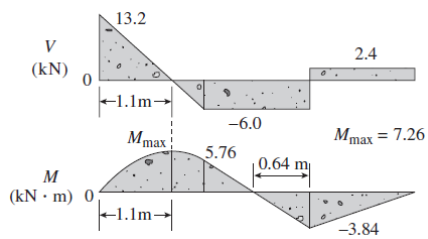
Date: 30 May, 2023

Prob. 1: At full draw, an archer applies a pull of 130 N to the bowstring of the bow shown in the figure. Determine the bending moment at the midpoint of the bow. (Ans: M=108 N.m)



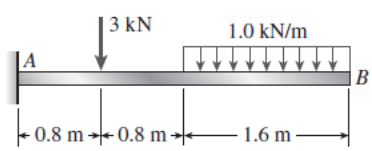
Prob. 2: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.

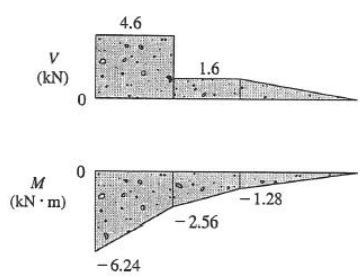


(Answer: 

)

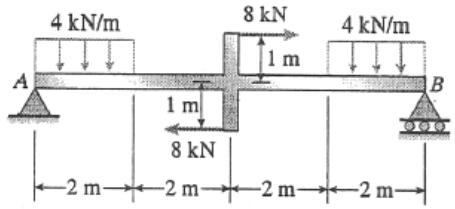
Prob. 3: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.

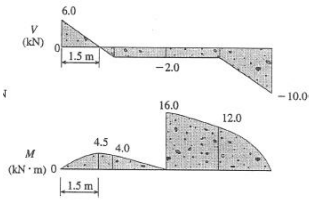


(Answer: 

)

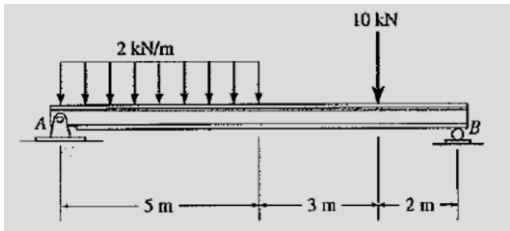
Prob. 4: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.



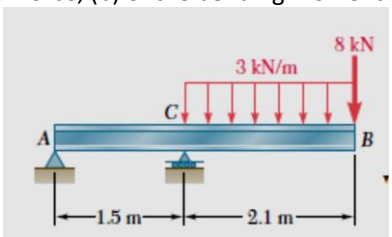
(Answer: 

)

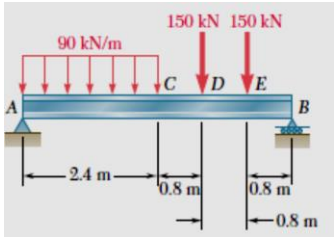
Prob. 5: Draw the shear force and bending moment diagram of the beam.



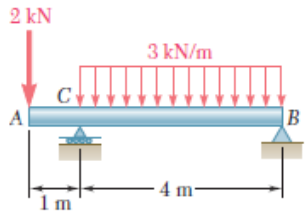
Prob. 6: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.



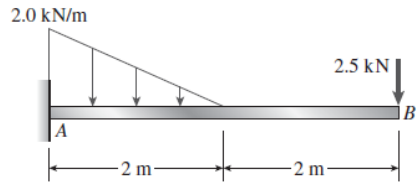
Prob. 7: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.

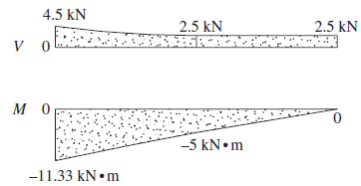


Prob. 8: Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear force, (b) of the bending moment.



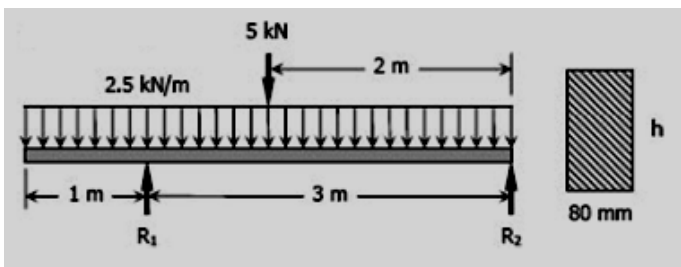
Prob. 9: The cantilever beam AB shown in the figure is subjected to a triangular load acting throughout one-half of its length and a concentrated load acting at the free end. Draw the shear-force and bending-moment diagrams for this beam.



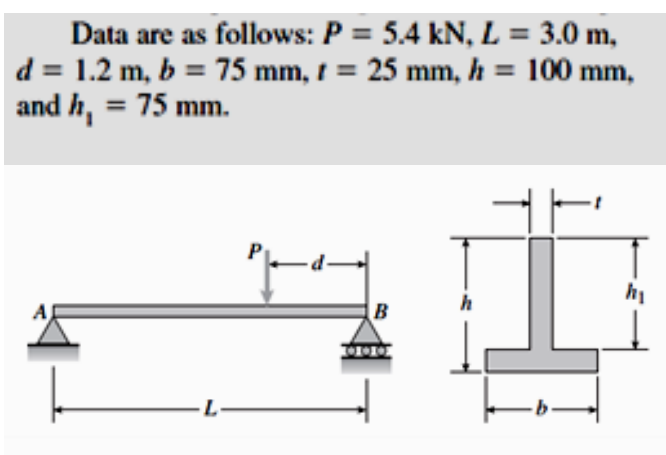
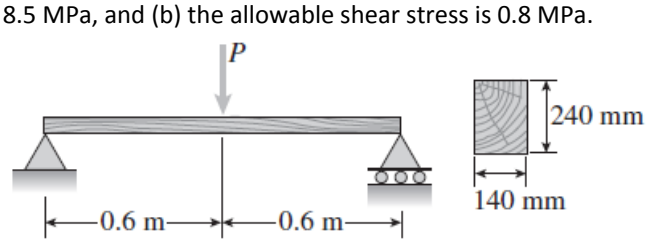
(Answer: 

)

Prob. 10: Determine the minimum height h of the beam shown in figure if the flexural stress is not to exceed 20MPa.



Prob. 11: Determine the maximum tensile stress σt and maximum compressive stress σc due to the load P acting on the simple beam AB.



Prob. 12: A simply supported wood beam of rectangular cross section and span length 1.2 m carries a concentrated load P at midspan in addition to its own weight (see figure). The cross section has width 140 mm and height 240 mm. The weight density of the wood is 5.4 kN/m3. Calculate the maximum permissible value of the load P if (a) the allowable bending stress is 8.5 MPa, and (b) the allowable shear stress is 0.8 MPa.